Virtual University of Pakistan

BIO203

Essentials of Genetics

Midterm past papers solved

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Bio203 24 dec 2018

Aflp pcr allel specific pcr5
Ans: AFLP PCR - METHODOLOGY
 Genomic DNA is digested with one or more restriction enzymes. tetracutter (MseI) and a hexacutter (EcoRI).

 Ligation of linkers to all restriction fragments. Pre-selective PCR is performed using primers which match the linkers and restriction site specific sequences.

 Electrophoretic separation and amplicons on a gel matrix, followed by visualization of the band pattern.

 AFLP is a highly sensitive PCR-based method for detecting polymorphisms in DNA.

 AFLP can be also used for genotyping individuals for a large number of loci.

ALLELE SPECIFIC PCR

 Allele-specific PCR used for identifying of SNPs.

 It requires prior knowledge of a DNA sequence, including differences between alleles.

 Uses primers whose 3' ends encompass the SNP.

 PCR amplification under stringent conditions is much less efficient in the presence of a mismatch between template and primer.

This diagnostic or cloning technique is used to identify or utilize singlenucleotide Polymorphisms (SNPs).

Mini primer pcr application polymerase name methadology 5

Ans: Mini Primer PCR uses a thermostable polymerase (S-Tbr) that can extend from short primers as short as 9 or 10 nucleotides.

Alu pcr 2

Ans: The use of primers from a commonly repeated segment is called Alu-PCR, and can help to amplify sequences adjacent (or between) these repeats.

Wga pcr3

Ans: Whole Genome Amplification (WGA) is a method for robust amplification of an entire genome, starting with nanogram quantities of DNA and resulting in microgram quantities of amplified products. WGA has become an invaluable method for preserving limited samples of precious stock material, particularly when using WGA methods that have been developed to amplify material from a single cell

Pfu and taq polymerase source extension rate 3

Ans:

- Pfu extention rate 60 Source *Pyrococcus furiosus*
- Taq pol extension rate 75 Source *T. aquaticus*

Bio203
timing: 10:30
My paper
1) multiplex PCR enables the simultaneous amplification in many targets on interest in ……
One reaction
2) Pcr amplification specificity is ensured by --------**(polymerase enzyme)**
3) to clone large genes not possible with conventional PCR.(Long pcr)
4) Allele-specific PCR used for identifying of **SNPs**.
5) **In Situ PCR (ISH)** is a polymerase chain reaction that actually takes place inside the cell on a slide.
6) Assembly PCR is the synthesis of long DNA structures by performing PCR on a pool of long **oligonucleotides** .
7) In sucide is typically used in **polegenetics**.
8) PCR used to produce a unique fingerprints of amplified product lengths is called **(INTER SEQUENCE SPECIFIC PCR)**
9) RT pcr is widely used in ----**(expression profiling).**
10) in nested pcr -----used to amplify a fragment**.(.two pairs of primer).**
11) in molecular technique the RAPD stand for**(.Random Amplified Polymeric DNa).**
12) Denaturation temperature:Reduces double stranded molecules to single stranded**, 90–96oC, 20-45 seconds.**
13) primer may occur due to the Hybridization of primer each other.
14) Taq polymerase most commonly used polymerase and code for **(Thermus aqauticus)..**
15) WGA is teachnique in which amplifications proceed ___in genome**.(Directly at any location)**
16) In annealing temp is lowered to ___ for 20 - 40 sec.**(40 - 65)**
17) PRIMERS: Primers are single-stranded **18–30 bp** long.
18) ..... Controls for contamination contains all reagents except DNA template**.(Blank reaction).**

19) Source of **Taq polerases** the Thermus aquaticus
20) Set of each primers work on the A**. high temp**

21) What is WGA technique

Ans: Whole Genome Amplification (WGA) is a method for robust amplification of an entire genome, starting with nanogram quantities of DNA and resulting in microgram quantities of amplified products. WGA has become an invaluable method for preserving limited samples of precious stock material, particularly when using WGA methods that have been developed to amplify material from a single cell

22) RAP stands for?

Ans:

23) use of methylation in pcr

Ans: Methylation-specific PCR is used to identify patterns of DNA methylation at CpG islands in genomic DNA. Target DNA is first treated with sodium bisulfite, which converts unmethylated cytosine bases to uracil, which is complementary to adenosine in PCR primers.

 Two amplifications are then carried out on the bisulfite-treated DNA: One primer set anneals to DNA with cytosines (corresponding to methylated cytosine), and the other set anneals to DNA with uracil (corresponding to unmethylated cytosine). MSP used in quantitative PCR provides information about methylation state of a given CpG island.

24) chemical ragents which is use in conventional PCR and also give reson why thy are use

Ans:
DNA template
 Primers
 Buffer
 MgCl2
 Deoxynucleotide triphosphates
 DNA polymerase

 Distilled water

25) Write the extention rate and source of Taq. Polymerase Vent and Pfu

Ans: 1) Taq pol extension rate 75 Source *T. aquaticus*

2) Vent	extension rate >80	Source *Thermococcus Litoralis*
3) Pfu	extention rate 60	Source *Pyrococcus furiosus*

26) Differeniate the methodology and principles of Allele specific and AFLP PCR

Ans: AFLP PCR - METHODOLOGY

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Bio203.

28-6-2018.

10:30am

Qno.1.steps of ligation mediated PCR?

Ans: Uses small DNA oligonucleotide 'linkers' (or adaptors) that are first ligated to fragments of the target DNA.

 PCR primers that anneal to the linker sequences are then used to amplify the target fragments.

 DNA sequencing

 Genome walking

 DNA footprinting

Qno.2.uses of methylation specific PCR?

Ans: Answer above

Qno.3.negative control and blank reaction ?

Ans: NEGATIVE CONTROL
 Controls for specificity of the amplification reaction contains all reagents and a DNA template lacking the target sequence.

BLANK REACTION
 Controls for contamination contains all reagents except DNA template.

Qno.4. write the extension rate and source of taq polymerase,pfu,vent?

Ans: Answer above

Qno.5. Write the chemical reagents used in PCR.give reason why they are used?

Ans: Answer above

Qno.6. methodology and principle of allelic specific PCR and AFLP PCR ?

Ans: Answer above
All objective from past papers.

Bio203

1) **What is methylation pcr**

Ans: Answer above

2) chemical reagents which is use in conventional PCR and also give reason why they are use

Ans: Answer above

3) Write the extension rate and source of Taq. Polymerase Vent and Pfu

Ans: Answer above

4) Differeniate the methodology and principles of Allele specific and AFLP PCR

Ans: Answer above

5) FIRST STEP OF RT PCR

Ans: REVERSE TRANSCRIPTASE PCR

 First step of RT-PCR - first strand reaction

 Synthesis of cDNA using oligo dT primers (37°C) one hour.

 Second strand reaction - digestion of cDNA:RNA hybrid (RNaseH)-

 Standard PCR with DNA oligo primers .

6) the name and length of polymerase used mini PCR

Ans: Mini Primer PCR uses a thermostable polymerase (S-Tbr) that can extend from short primers as short as 9 or 10 nucleotides.

MCQ
1) Multiplex PCR enables the simultaneous amplification in many targets on interest in …… ans) **One reaction**
2) RT pcr is widely used in **(expression profiling).**
3) . Controls for contamination contains all reagents except DNA template.**(Blank reaction).**
4) 2) Denaturation temperature: Reduces double stranded molecules to single stranded, **90–96oC, 20-45 seconds**.
5) in nested pcr -----used to amplify a fragment.**(.two pairs of primer).**
6) PRIMERS: Primers are single-stranded **18–30 bp** long.
7) LONG PCR - to clone**(large genes not)** possible with conventional PCR
😎🤓 SOURTH BLOTING USED TO DETECT
9) PCR can be perfom to amlify
10) TAq polymerase optimum activity tem **70 to 80 c**

Bio203, timing 5:30
What is WAG technique
use of methylation in pcr
chemical ragents which is use in conventional PCR and also give reson why thy are used
Differeniate the methodology and principles of Allele specific and AFLP PCR

Principle of ligated mediated Pcr
Negative control and blank reaction

Write 1st step of RT-PCR(2)
Uses of methylation PCR(2)
Descib denaturation step of PCR(3)

Ans: First regular cycling event and consists of heating the reaction to 93°C - 98°C for 20-45 seconds.

 It causes melting of DNA template yielding single strands of DNA.

Denaturation temperature
 Reduces double stranded molecules to single stranded

 90–96oC, 20-45 seconds

ALU-PCR(3)

Ans: The use of primers from a commonly repeated segment is called Alu-PCR, and can help to amplify sequences adjacent (or between) these repeats.

Mini-primers(5)
Chemical reagents of PCR and their uses?..(5)

BIO203
23 June 2018. 12:00 pm
20 mcq's + 6 questions

1) how template is different in RT-PCR from conventional PCR?

Ans: In conventional PCR the template is DNA while in RT-PCR the template we use is RNA.

2) methodology of QRT-PCR.

Ans: QRT-PCR methods use fluorescent dyes, such as Sybr Green, or fluorophore-containing DNA probes, such as TaqMan, to measure the amount of amplified product in real time.

3) write first step of RT-PCR.

Ans: Assembly PCR is the synthesis of long DNA structures by performing PCR on a pool of long oligonucleotides with short overlapping segments, to assemble two or more pieces of DNA into one piece.

4) write methodology of assembly PCR.

Ans: It involves an initial PCR with primers that have an overlap and a second PCR using the products as the template that generates the final full-length product.

5) write the names of chemical reagents used in standard PCR and also write their uses.

Ans: Answer above mentioned.

6) differentiate between methodology and application of ALFP-PCR and Allele Specific PCR.

Ans: Answer above mentioned

----------GOOD LUCK----------